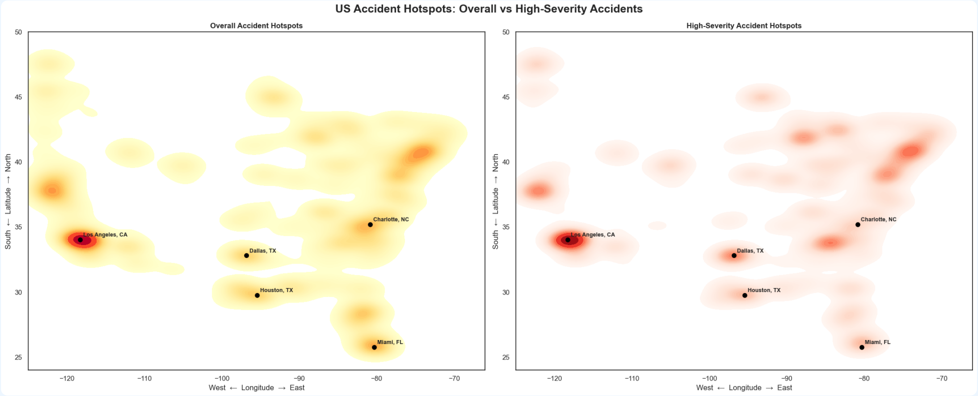
US Accidents

Exploratory Data Analysis

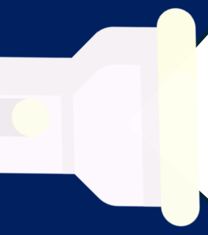
2016–2023 | Patterns in severity, time, weather, road features & hotspots



Source: Kaggle US Accidents dataset

What this analysis answers

The notebook turns a large accident dataset into four practical questions.



Severity

Which severity level is most common?

Weather

Which conditions are associated with higher severity?

Time

When do accidents happen?

Road context

Which features and locations signal risk?

Most reported accidents are Severity 2

The dataset contains 7.73M records; lower-to-moderate severity dominates the distribution.

7.73M

total accident records

19.5%

Severity 3–4 share

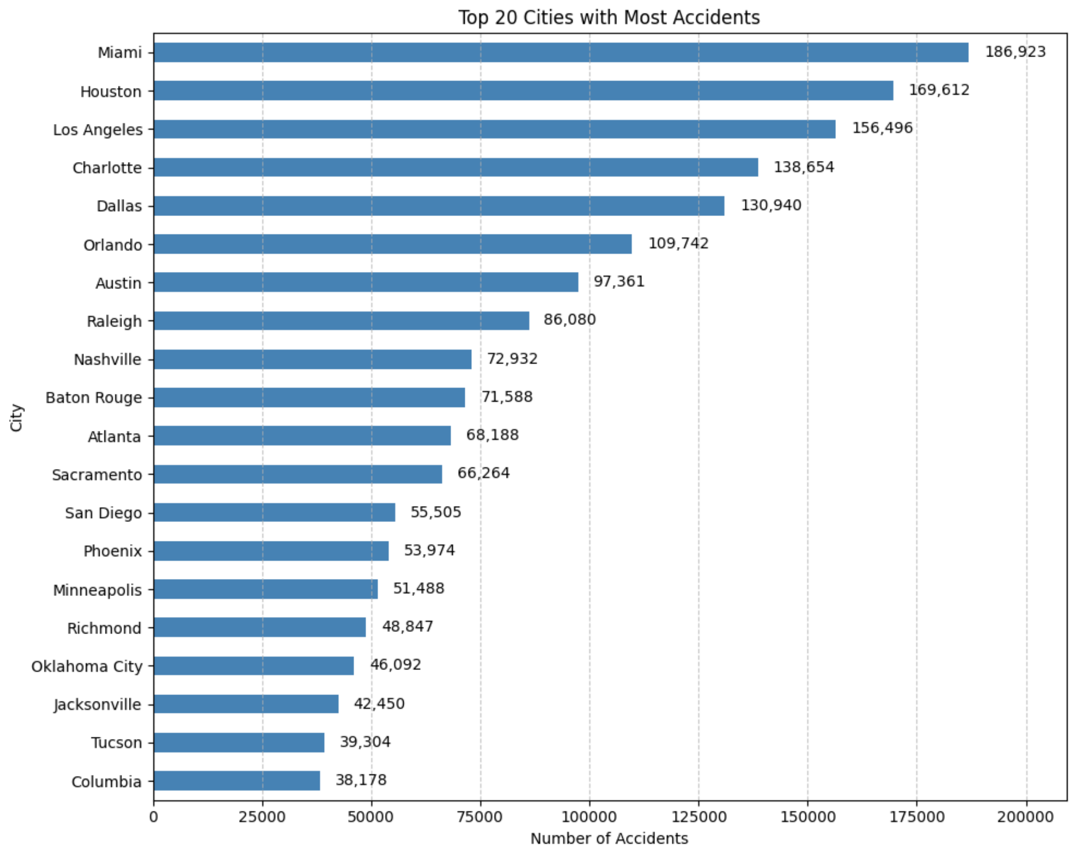
79.7%

Severity 2 share

Severity counts: 1 = 67,366 · 2 = 6,156,981 · 3 = 1,299,337 · 4 = 204,710

Accident volume is concentrated in major metro areas

Miami appears as the largest city in the notebook output, followed by Houston and Los Angeles.



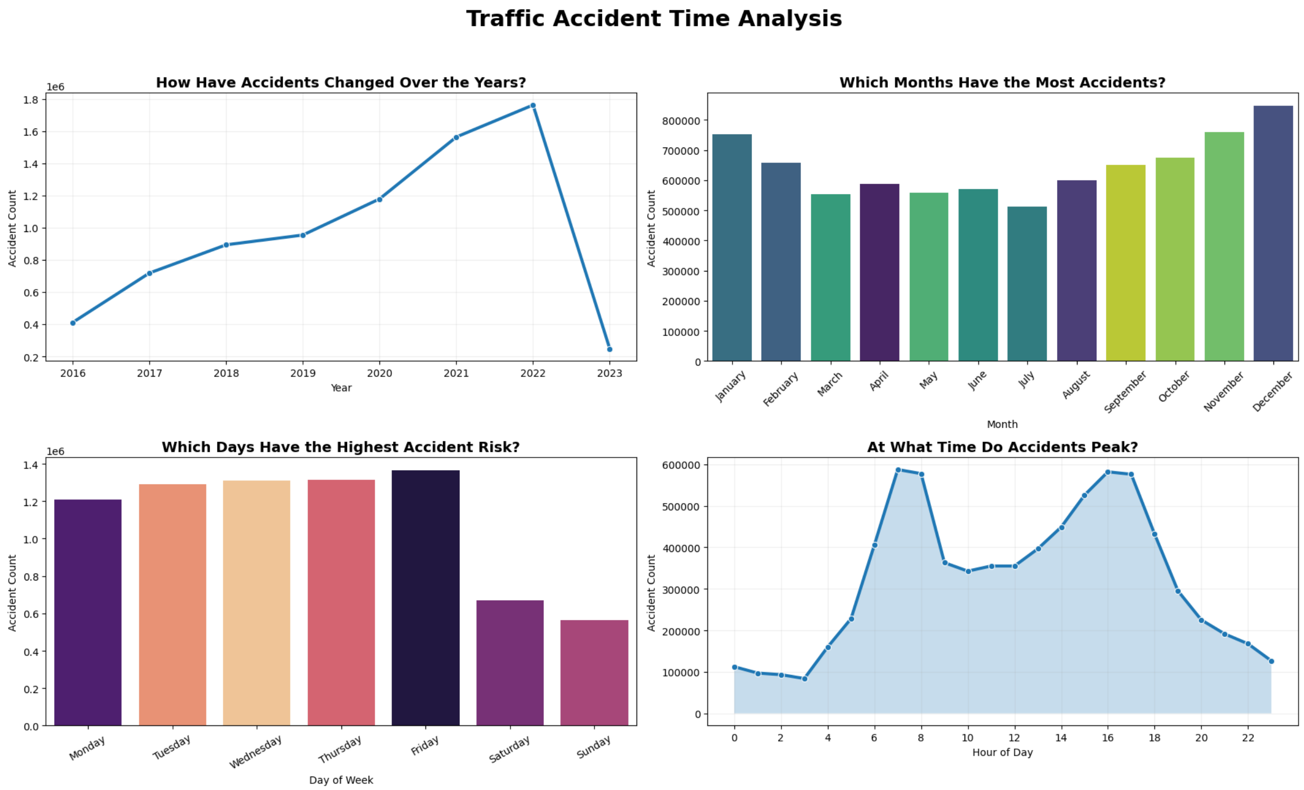
186,923

accidents in Miami (the highest-count city)

Target urban corridors with location-specific safety interventions and operational monitoring.

Accidents vary by year, season, weekday and hour

Time analysis is the strongest starting point for staffing, alerts and prevention planning.



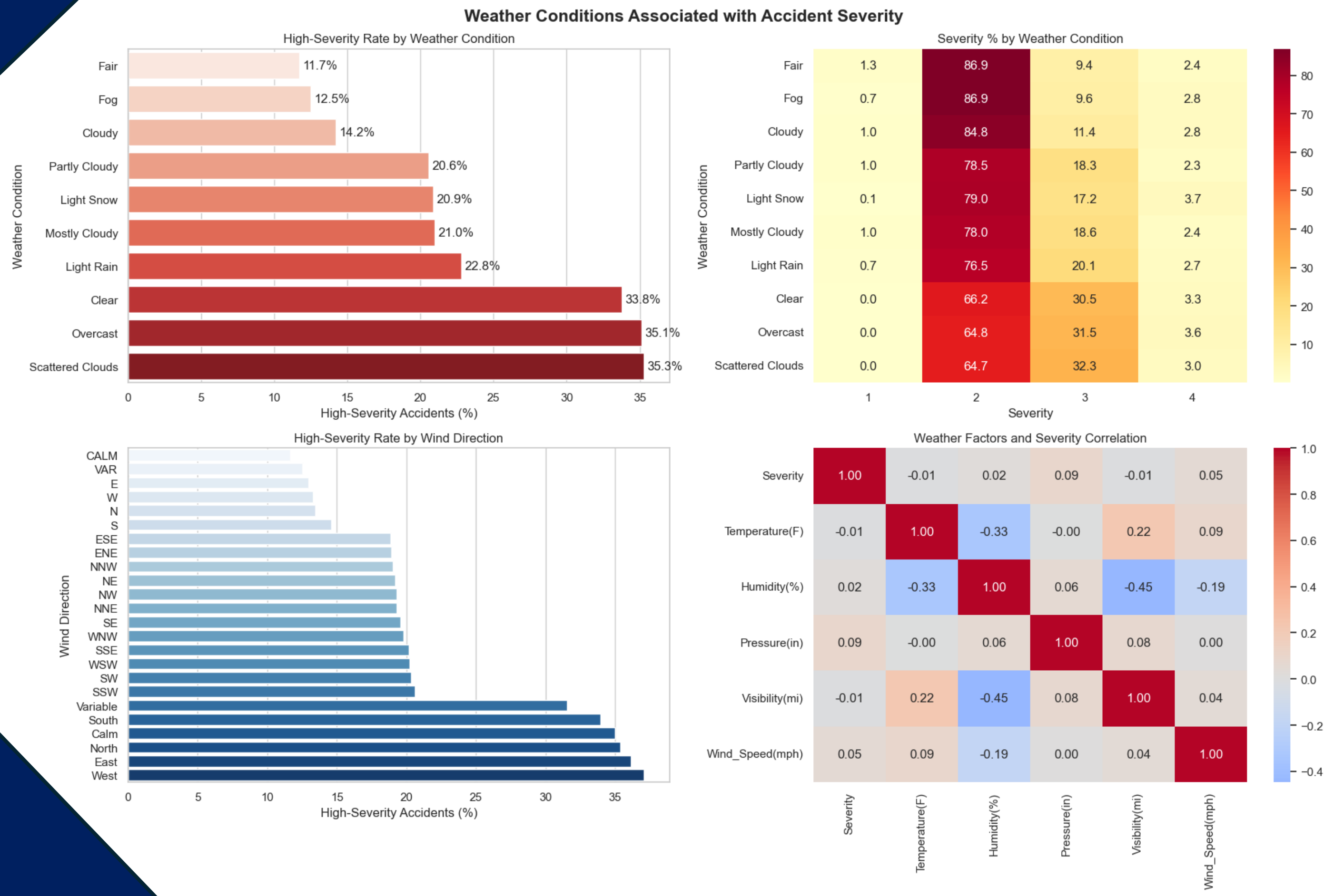
Use the dashboard to isolate peaks by local market.

The notebook covers four lenses:

- Year trend
- Monthly seasonality
- Day-of-week pattern
- Hourly pattern

Weather alone is not a complete explanation for severity

The notebook compares high-severity rates, severity mix, wind direction and weather correlations.

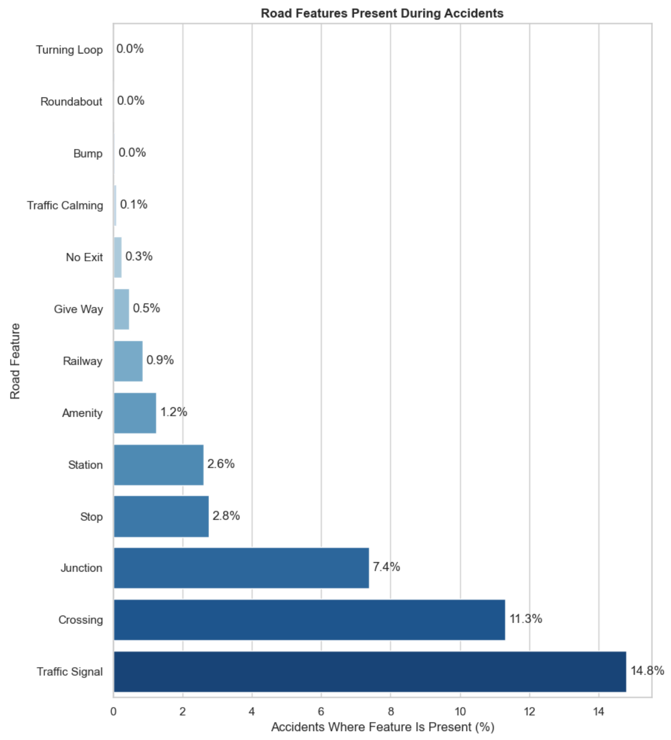


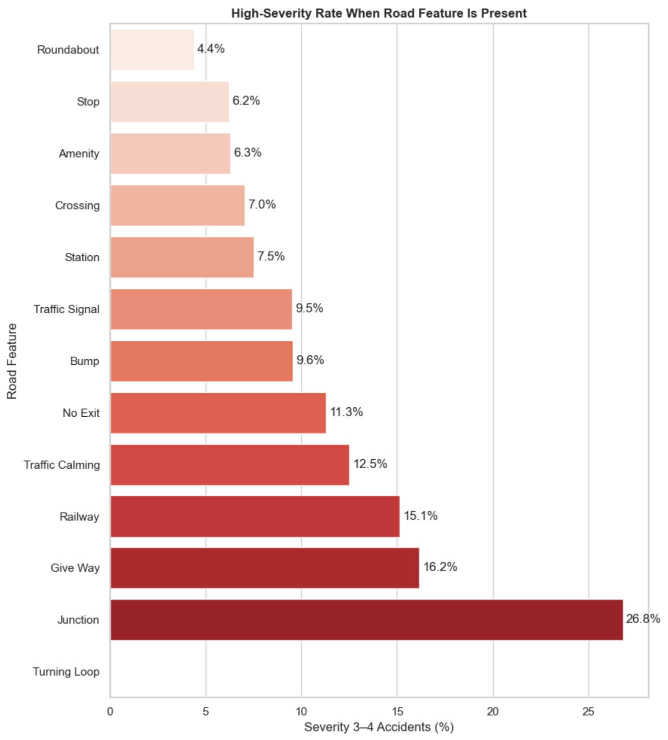
Interpret carefully

Correlation helps screen factors, but it does not prove weather caused an accident. Pair weather signals with road, traffic and location context.

Junctions stand out as a high-severity road context

Traffic signals and crossings are common at accident sites; junctions have the strongest reported high-severity rate.





14.8%

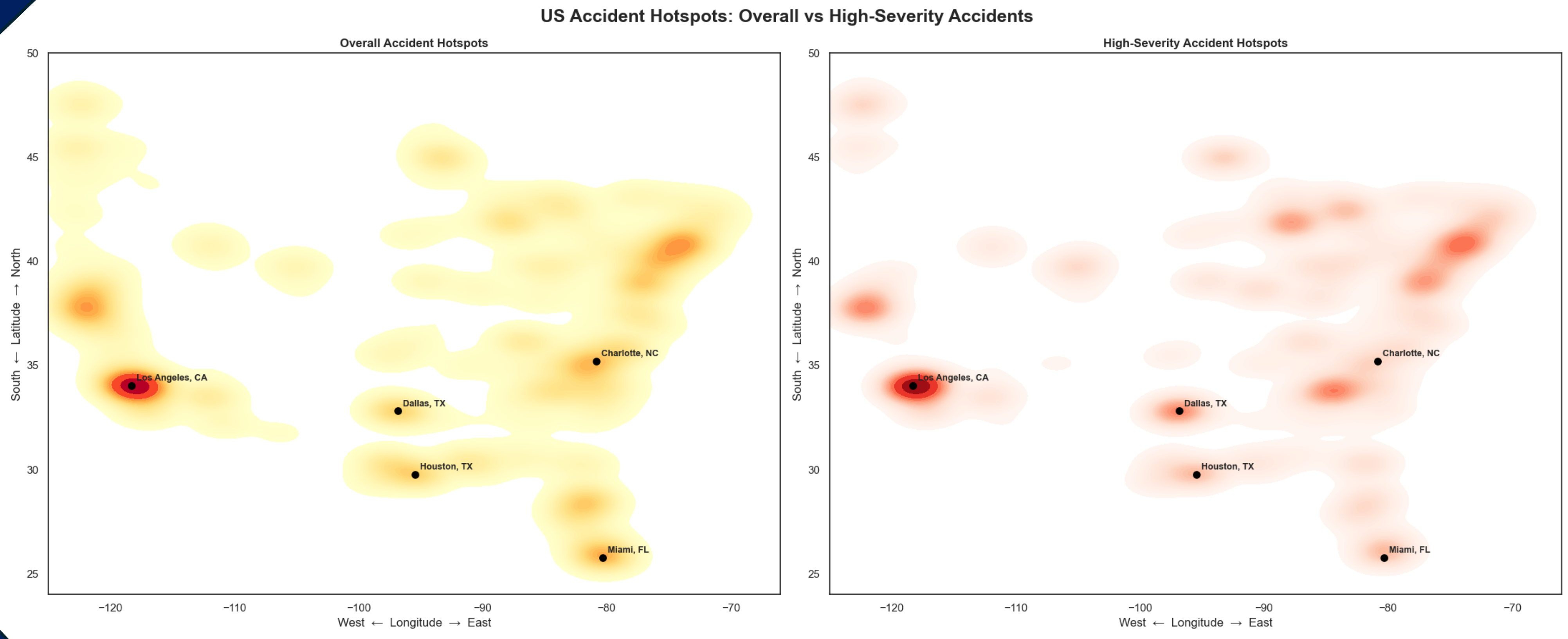
accidents near traffic signals

26.8%

high-severity rate at junctions

Hotspots separate overall volume from severe risk

Comparing all accidents with Severity 3–4 events helps prioritize interventions beyond raw volume.



Key takeaways for a safety-focused dashboard

Prioritize metro hotspots

Start with high-volume cities and state corridors.

Plan around time patterns

Use year/month/day/hour filters for local operations.

Flag junction risk

Monitor junctions where high-severity share is elevated.

Use weather as context

Combine conditions with traffic, road features and geography.